Trade-offs between unitary and measurement induced spin squeezing in cavity QED: Supplementary Material

In this Supplementary Material we show how to derive Eq. (1) starting from the fundamental atom-light interaction inside a cavity through adiabatic elimination of excited optical states and cavity degrees of freedom. We then explain how, starting from Eq. (1), we can obtain Eq. (2) and Eq. (3) for second and first order correlators, respectively, by using the large N and gaussianity assumptions. We continue by showing how these correlators are related to quantities of interest like spin squeezing and state area. We then portray how to obtain the optimal spin squeezing and state area in the setting of QND measurements, both in the presence and absence of spin flips. We finalize by repeating the same calculations for OAT dynamics.

EFFECTIVE EVOLUTION EQUATION

Here we show how to obtain Eq. (1) starting from the fundamental atom-light interaction inside a cavity. We consider N three-level atoms (levels $|e\rangle$, $|{\downarrow}\rangle$, $|{\uparrow}\rangle$ with energies ω_e, $\omega_\downarrow$ and $\omega_\uparrow$) in the configuration shown in Fig. 1(a). The atoms interact with one cavity mode at frequency ω_c, with single photon Rabi frequencies $2g_1$ and $2g_2$ for the transitions $|e\rangle \leftrightarrow |{\uparrow}\rangle$ and $|e\rangle \leftrightarrow |{\downarrow}\rangle$, respectively. The cavity has power decay linewidth κ, and the atoms spontaneously emit from $|e\rangle \to |{\uparrow}\rangle$ and from $|e\rangle \to |{\downarrow}\rangle$ with rates γ_1 and γ_2, respectively. Furthermore, the cavity is externally driven by a laser at frequency ω_d (close to ω_c) and with a flux of $|\beta|^2$ photons per second. The light escaping the cavity is then collected and subjected to homodyne detection with overall efficiency η. The evolution of the combined atom-light system in the presence of all these processes is described by the following stochastic differential equation

$$
\begin{aligned}
d\hat{\rho} = &-i\Big[\underbrace{\omega_e\hat{N}_e+\omega_\uparrow\hat{N}_\uparrow+\omega_\downarrow\hat{N}_\downarrow+\omega_c\hat{a}^\dagger\hat{a}}_{\hat{H}_{\text{free}}}+\underbrace{g_1(\hat{a}\hat{S}_{e\uparrow}+\text{h.c.})+g_2(\hat{a}\hat{S}_{e\downarrow}+\text{h.c.})}_{\hat{H}_{\text{int}}}+\underbrace{\beta\sqrt{\kappa}(\hat{a}e^{i\omega_d t}+\text{h.c.})}_{\hat{H}_{\text{drive}}},\hat{\rho}\Big]\,dt\\
&+\underbrace{\kappa\Big(\hat{a}\hat{\rho}\hat{a}^\dagger-\frac{\{\hat{a}^\dagger\hat{a},\hat{\rho}\}}{2}\Big)}_{\text{photon leakage}}dt+\underbrace{\gamma_1\sum_{i=1}^{N}\Big(\hat{\sigma}^i_{\uparrow e}\,\hat{\rho}\,\hat{\sigma}^i_{e\uparrow}-\frac{\{\hat{\sigma}^i_{ee},\hat{\rho}\}}{2}\Big)dt+\gamma_2\sum_{i=1}^{N}\Big(\hat{\sigma}^i_{\downarrow e}\,\hat{\rho}\,\hat{\sigma}^i_{e\downarrow}-\frac{\{\hat{\sigma}^i_{ee},\hat{\rho}\}}{2}\Big)dt}_{\text{spontaneous emission}}\\
&+\underbrace{\sqrt{\kappa\eta}\Big(\hat{a}e^{-i\phi}\hat{\rho}+\hat{\rho}\hat{a}^\dagger e^{i\phi}-\langle\hat{a}e^{-i\phi}+\hat{a}^\dagger e^{i\phi}\rangle\hat{\rho}\Big)}_{\text{homodyne detection}}\,dW,
\end{aligned}
\tag{S1}
$$

where $\hat{\rho}$ is the density matrix of the atom-light system, $\hat{\sigma}^i_{ab} = |a\rangle\langle b|_i$ $(a, b = e, \uparrow, \downarrow)$ is a single particle transition operator for atom i, $\hat{S}_{ab} = \sum_{i=1}^N \hat{\sigma}^i_{ab}$ is a collective transition operator, $\hat{N}_a = \hat{S}_{aa}$ is the "number of atoms in level a" operator, $\hat{a}^\dagger$ ($\hat{a}$) are bosonic creation (annihilation) operators for the cavity mode and ϕ is the homodyne detection angle, which is an experimentally adjustable parameter. The first line describes coherent evolution under a Hamiltonian that consists of three parts: the free uncoupled evolution of atoms and cavity, the atom-light interaction (of Jaynes-Cummings type), and the external laser drive. The second line describes the dissipative processes: photon leakage and spontaneous emission from the excited state. The third line implements homodyne detection [S1, S2], which is intimately tied to photon leakage since these photons are the ones that are being measured.

Since the cavity is detuned from the $|{\downarrow}\rangle, |{\uparrow}\rangle \leftrightarrow |e\rangle$ by an amount $\pm\Delta$, the atom-light interaction becomes dispersive when $|\Delta| \gg g_{1,2}\sqrt{N}, g_{1,2}\sqrt{n_{\text{phot}}}$, where n_{phot} is the number of photons inside the cavity. We implement this mathematically by means of a Schrieffer-Wolff transformation. To be more precise, we define a dressed density matrix $\hat{\rho}_d = e^{\hat{W}}\,\hat{\rho}\,e^{-\hat{W}}$, where

$$
\hat{W} = \frac{g_1}{\Delta}(\hat{a}^\dagger\hat{S}_{\uparrow e} - \hat{a}\hat{S}_{e\uparrow}) + \frac{g_2}{\Delta}(\hat{a}\hat{S}_{e\downarrow} - \hat{a}^\dagger\hat{S}_{\downarrow e})
\tag{S2}
$$

is the Schrieffer-Wolff generator (note $e^{\hat{W}}$ is unitary). The conditions $|\Delta| \gg g_{1,2}\sqrt{N}, g_{1,2}\sqrt{n_{\text{phot}}}$ then mean that $\hat{W}$ acts perturbatively in the states of interest. The evolution equation for $\hat{\rho}_d$ can then be written down by considering the following relations:

$$
\begin{aligned}
e^{\hat{W}}(\hat{H}_{\text{free}}+\hat{H}_{\text{int}})e^{-\hat{W}} &= \hat{H}_{\text{free}}+\frac{g_1^2}{\Delta}[\hat{a}^\dagger\hat{a}(\hat{S}_{\uparrow\uparrow}-\hat{S}_{ee})-\hat{S}_{e\uparrow}\hat{S}_{\uparrow e}]-\frac{g_2^2}{\Delta}[\hat{a}^\dagger\hat{a}(\hat{S}_{\downarrow\downarrow}-\hat{S}_{ee})-\hat{S}_{e\downarrow}\hat{S}_{\downarrow e}]+O(g_{1,2}^3/\Delta^2)\\
e^{\hat{W}}\,\hat{a}\,e^{-\hat{W}} &= \hat{a}+\frac{1}{\Delta}(g_2\hat{S}_{\downarrow e}-g_1\hat{S}_{\uparrow e})+O(g_{1,2}^2/\Delta^2)\\
e^{\hat{W}}\,\hat{\sigma}^i_{\uparrow e}\,e^{-\hat{W}} &= \hat{\sigma}^i_{\uparrow e}+\frac{g_1\hat{a}}{\Delta}(\hat{\sigma}^i_{\uparrow\uparrow}-\hat{\sigma}^i_{ee})-\frac{g_2\hat{a}}{\Delta}\hat{\sigma}^i_{\uparrow\downarrow}+O(g_{1,2}^2/\Delta^2)\\
e^{\hat{W}}\,\hat{\sigma}^i_{\downarrow e}\,e^{-\hat{W}} &= \hat{\sigma}^i_{\uparrow e}+\frac{g_2\hat{a}}{\Delta}(\hat{\sigma}^i_{\downarrow\downarrow}-\hat{\sigma}^i_{ee})+\frac{g_1\hat{a}}{\Delta}\hat{\sigma}^i_{\downarrow\uparrow}+O(g_{1,2}^2/\Delta^2)\\
e^{\hat{W}}\,\hat{\sigma}^i_{ee}\,e^{-\hat{W}} &= \hat{\sigma}^i_{ee}+\frac{1}{\Delta}\Big[\hat{a}(g_1\hat{\sigma}^i_{e\uparrow}-g_2\hat{\sigma}^i_{e\downarrow})+\hat{a}^\dagger(g_1\hat{\sigma}^i_{\uparrow e}-g_2\hat{\sigma}^i_{\downarrow e})\Big]\\
&\quad+\frac{\hat{a}^\dagger\hat{a}}{\Delta}\Big[g_1^2(\hat{\sigma}^i_{\uparrow\uparrow}-\hat{\sigma}^i_{ee})+g_2^2(\hat{\sigma}^i_{\downarrow\downarrow}-\hat{\sigma}^i_{ee})-g_1g_2(\hat{\sigma}^i_{\uparrow\downarrow}+\hat{\sigma}^i_{\downarrow\uparrow})\Big]\\
&\quad+\frac{1}{2\Delta}\Big[g_1g_2(\hat{\sigma}^i_{e\downarrow}\hat{S}_{\uparrow e}+\hat{\sigma}^i_{e\uparrow}\hat{S}_{\downarrow e})-g_1^2\hat{\sigma}^i_{e\uparrow}\hat{S}_{\uparrow e}-g_2^2\hat{\sigma}^i_{e\downarrow}\hat{S}_{\downarrow e}+\text{ h.c.}\Big]+O(g_{1,2}^3/\Delta^3)
\end{aligned}
\tag{S3}
$$

We have expanded the transformed operators only to the order that will be relevant for the evolution equation of $\hat{\rho}_d$. However, before writing this equation down we will point out a few features that will allow us to simplify the result. First, single particle decoherence will now include terms of the form $\gamma_{1,2}g_1\hat{\sigma}^i_{e\uparrow}\hat{a}\hat{\rho}_d/\Delta, \gamma_{1,2}g_2\hat{\sigma}^i_{e\downarrow}\hat{a}\hat{\rho}_d/\Delta$ (and complex conjugates), which transfer atoms from the ground manifold to the excited state. However, they oscillate at frequency $\pm\Delta$ (due to $\hat{H}_{\text{free}}$) and can thus be neglected since they will effectively contribute corrections of size $\gamma_{1,2}^2g^2/\Delta^3$ to the evolution equation. These terms become relevant when $\Delta \leq \gamma$ in which case a more refined treatment is required where both spontaneous emission and $\hat{H}_{\text{free}}$ are included as the zeroth order term of a dissipative Schrieffer-Wolff transformation. We do not attempt this since it is considerably more challenging to implement from a technical standpoint and is outside our regime of interest anyway, but this means that $\Delta \gtrsim \gamma$ is another assumption of our treatment. After neglecting these terms, the resulting equation will preserve the $|\uparrow\rangle, |\downarrow\rangle$ manifold. In the dressed basis, all atoms are in the ground states, and any admixture with $|e\rangle$ is taken care of by the dressing, which allows us to get rid of any term of the form $\hat{\sigma}^i_{\uparrow e}\hat{\rho}$, $\hat{\sigma}^i_{\downarrow e}\hat{\rho}$, $\hat{\sigma}^i_{ee}\hat{\rho}$ (and their complex conjugates). All these simplifications are generic after adiabatic elimination of $|e\rangle$. However, in the specific case we are considering, there is also a large energy difference between $|\uparrow\rangle$ and $|\downarrow\rangle$ of $\pm 2\Delta$. This means that terms of the form $\hat{\sigma}^i_{\uparrow\downarrow}\hat{\rho}_d\hat{\sigma}^i_{\uparrow\uparrow}$, $\hat{\sigma}^i_{\uparrow\downarrow}\hat{\rho}_d$ (and similar), which arise due to interfering decay pathways, oscillate very fast and can be neglected too. Taking all of this into consideration, the resulting equation for $\hat{\rho}_d$ is

$$
\begin{aligned}
d\hat{\rho}_d = &-i\Big[\omega_c\hat{a}^\dagger\hat{a}+\omega_\downarrow\hat{N}_\downarrow+\omega_\uparrow\hat{N}_\uparrow+\frac{(g_1^2-g_2^2)}{2\Delta}\hat{a}^\dagger\hat{a}\,N+\frac{g_1^2+g_2^2}{2\Delta}\hat{a}^\dagger\hat{a}\,(\hat{S}_{\uparrow\uparrow}-\hat{S}_{\downarrow\downarrow})+\beta\sqrt{\kappa}(\hat{a}e^{i\omega_d t}+\text{h.c.}),\hat{\rho}_d\Big]\,dt\\
&+\frac{\gamma_1g_1^2}{\Delta^2}\sum_{i=1}^N\Big(\hat{\sigma}^i_{\uparrow\uparrow}\hat{a}\,\hat{\rho}_d\,\hat{a}^\dagger\hat{\sigma}^i_{\uparrow\uparrow}-\frac{\{\hat{a}^\dagger\hat{a}\,\hat{\sigma}^i_{\uparrow\uparrow},\hat{\rho}_d\}}{2}\Big)\,dt+\frac{\gamma_2g_2^2}{\Delta^2}\sum_{i=1}^N\Big(\hat{\sigma}^i_{\downarrow\downarrow}\hat{a}\,\hat{\rho}_d\,\hat{a}^\dagger\hat{\sigma}^i_{\downarrow\downarrow}-\frac{\{\hat{a}^\dagger\hat{a}\,\hat{\sigma}^i_{\downarrow\downarrow},\hat{\rho}_d\}}{2}\Big)\,dt\\
&+\frac{\gamma_1g_2^2}{\Delta^2}\sum_{i=1}^N\Big(\hat{\sigma}^i_{\uparrow\downarrow}\hat{a}\,\hat{\rho}_d\,\hat{a}^\dagger\hat{\sigma}^i_{\downarrow\uparrow}-\frac{\{\hat{a}^\dagger\hat{a}\,\hat{\sigma}^i_{\downarrow\downarrow},\hat{\rho}_d\}}{2}\Big)\,dt+\frac{\gamma_2g_1^2}{\Delta^2}\sum_{i=1}^N\Big(\hat{\sigma}^i_{\downarrow\uparrow}\hat{a}\,\hat{\rho}_d\,\hat{a}^\dagger\hat{\sigma}^i_{\uparrow\downarrow}-\frac{\{\hat{a}^\dagger\hat{a}\,\hat{\sigma}^i_{\uparrow\uparrow},\hat{\rho}_d\}}{2}\Big)\,dt\\
&+\kappa\Big(\hat{a}\hat{\rho}_d\hat{a}^\dagger-\frac{\{\hat{a}^\dagger\hat{a},\hat{\rho}\}}{2}\Big)\,dt+\sqrt{\kappa\eta}\Big(\hat{a}e^{-i\phi}\hat{\rho}_d+\hat{\rho}_d\hat{a}^\dagger e^{i\phi}-\langle\hat{a}e^{-i\phi}+\hat{a}^\dagger e^{i\phi}\rangle\hat{\rho}_d\Big)\,dW.
\end{aligned}
\tag{S4}
$$

Note that in the photon leakage, laser drive and homodyne detection terms, the leading order contribution is zeroth order in g/Δ. The single particle decoherence terms, which are $O(g^2/\Delta^2)$, describe off-resonant scattering of cavity light (hence the presence of the operators $\hat{a}$ and $\hat{a}^\dagger$) and involve both spin flip and spin conserving processes. We also now define $\hat{S}_z = (\hat{S}_{\uparrow\uparrow}-\hat{S}_{\downarrow\downarrow})/2$ and $\hat{N}_g = \hat{S}_{\uparrow\uparrow}+\hat{S}_{\downarrow\downarrow}$ for notational simplicity. Furthermore, since we are considering only the ground manifold, we will now write $\hat{\sigma}^i_{\uparrow\uparrow}+\hat{\sigma}^i_{\downarrow\downarrow}=1$.

The drive will establish a field inside the cavity that will depend on the drive-cavity detuning. However, the frequency of the cavity now depends on the state of the atoms, through the $\hat{a}^\dagger\hat{a}\,\hat{S}_z$ term, and will create an effective nonlinear atom-atom interaction. We take this into account perturbatively by analyzing fluctuations about the cavity field that would be established at the mean field level, taking into account that the average inversion is $\langle\hat{S}_z\rangle = 0$ for the initial conditions we want to consider [and it will remain so, as per Eq. (S4)]. We thus define the fluctuation

$\hat{b} = \hat{a} - \alpha e^{-i\omega_d t}$, where

$$\alpha = -\frac{i\sqrt{\kappa}\beta}{-i\delta_* + \kappa/2}, \tag{S5}$$

and $\delta_* = \omega_d - \omega_c - (g_1^2 - g_2^2)N/(2\Delta)$. At this point it is convenient to move into the rotating frame of the drive. In this frame, and in terms of the fluctuation field $\hat{b}$, we approximately get

$$\begin{aligned} d\hat{\rho}_d = & -i\Big[-\delta_* \hat{b}^\dagger \hat{b} + \frac{(\omega_\downarrow + \omega_\uparrow)}{2}\hat{N}_g + \Big(2\Delta + \frac{(g_1^2+g_2^2)|\alpha|^2}{\Delta}\Big)\hat{S}_z + \frac{(g_1^2+g_2^2)(\alpha\hat{b}^\dagger + \alpha^*\hat{b})\hat{S}_z}{\Delta}, \hat{\rho}_d\Big]\, dt \\ & + \kappa\Big(\hat{b}\hat{\rho}_d\hat{b}^\dagger - \frac{\{\hat{b}^\dagger\hat{b}, \hat{\rho}\}}{2}\Big)\, dt + \frac{(\gamma_1 g_1^2 + \gamma_2 g_2^2)|\alpha|^2}{2\Delta^2}\sum_{i=1}^N \Big(\hat{\sigma}^i_{\uparrow\uparrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\uparrow\uparrow} + \hat{\sigma}^i_{\downarrow\downarrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\downarrow\downarrow} - \hat{\rho}_d\Big)\, dt \\ & + \frac{\gamma_1 g_2^2|\alpha|^2}{\Delta^2}\sum_{i=1}^N \Big(\hat{\sigma}^i_{\uparrow\downarrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\downarrow\uparrow} - \frac{\{\hat{\sigma}^i_{\downarrow\downarrow}, \hat{\rho}_d\}}{2}\Big)\, dt + \frac{\gamma_2 g_1^2|\alpha|^2}{\Delta^2}\sum_{i=1}^N \Big(\hat{\sigma}^i_{\downarrow\uparrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\uparrow\downarrow} - \frac{\{\hat{\sigma}^i_{\uparrow\uparrow}, \hat{\rho}_d\}}{2}\Big)\, dt \\ & + \sqrt{\kappa\eta}\Big(\hat{b}e^{-i\phi}\hat{\rho}_d + \hat{\rho}_d\hat{b}^\dagger e^{i\phi} - \langle \hat{b}e^{-i\phi} + \hat{b}^\dagger e^{i\phi}\rangle\hat{\rho}_d\Big)\, dW. \end{aligned} \tag{S6}$$

In the previous equations we have symmetrized the spin conserving single particle processes by using the relation $\hat{\sigma}^i_{\uparrow\uparrow} + \hat{\sigma}^i_{\downarrow\downarrow} = 1$, kept only the leading order terms (in $|\alpha|$) in the single particle decoherence contributions, and we have neglected $(g_1^2 + g_2^2)\hat{b}^\dagger\hat{b}\,\hat{S}_z/\Delta$ in the Hamiltonian part, which is a nonlinear fluctuation term. It amounts to assuming that the quantum fluctuations in $\hat{b}$ that arise because of $\hat{S}_z$ are smaller than $|\alpha|$. We now adiabatically eliminate the cavity degree of freedom. This can be done by means of a dissipative Schrieffer-Wolff transformation (since κ and δ may be comparable now), but the fastest way to obtain the result is by doing the replacement

$$\hat{b} \to -\frac{i(g_1^2+g_2^2)\alpha\hat{S}_z/\Delta}{-i\delta_* + \kappa/2}, \tag{S7}$$

which is what a mean field treatment suggests will be the steady state intracavity field, on average. This leads to

$$\begin{aligned} d\hat{\rho}_d = & -i\Big[\frac{(\omega_\downarrow + \omega_\uparrow)}{2}\hat{N}_g + \Big(2\Delta + \frac{(g_1^2+g_2^2)|\beta|^2}{\Delta}\Big)\hat{S}_z + \Big(\frac{(g_1^2+g_2^2)|\alpha|}{\Delta}\Big)^2 \frac{\delta_*\,\hat{S}_z^2}{\delta_*^2 + (\kappa/2)^2}, \hat{\rho}_d\Big]\, dt \\ & + \Big(\frac{(g_1^2+g_2^2)|\alpha|}{\Delta}\Big)^2 \frac{\kappa}{\delta_*^2 + (\kappa/2)^2}\Big(\hat{S}_z\hat{\rho}_d\hat{S}_z - \frac{\{\hat{S}_z^2, \hat{\rho}\}}{2}\Big)\, dt + \frac{\gamma_1 g_2^2|\alpha|^2}{\Delta^2}\sum_{i=1}^N \Big(\hat{\sigma}^i_{\uparrow\downarrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\downarrow\uparrow} - \frac{\{\hat{\sigma}^i_{\downarrow\downarrow}, \hat{\rho}_d\}}{2}\Big)\, dt \\ & + \frac{\gamma_2 g_1^2|\alpha|^2}{\Delta^2}\sum_{i=1}^N \Big(\hat{\sigma}^i_{\downarrow\uparrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\uparrow\downarrow} - \frac{\{\hat{\sigma}^i_{\uparrow\uparrow}, \hat{\rho}_d\}}{2}\Big)\, dt + \frac{(\gamma_1 g_1^2 + \gamma_2 g_2^2)|\alpha|^2}{2\Delta^2}\sum_{i=1}^N \Big(\hat{\sigma}^i_{\uparrow\uparrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\uparrow\uparrow} + \hat{\sigma}^i_{\downarrow\downarrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\downarrow\downarrow} - \hat{\rho}_d\Big)\, dt \\ & + \sqrt{\Big(\frac{(g_1^2+g_2^2)|\alpha|}{\Delta}\Big)^2 \frac{\kappa\eta}{\delta_*^2 + (\kappa/2)^2}}\Big(\hat{S}_z\hat{\rho}_d + \hat{\rho}_d\hat{S}_z - 2\langle\hat{S}_z\rangle\hat{\rho}_d\Big)\, dW. \end{aligned} \tag{S8}$$

At this point it is convenient to introduce the parameters

$$\begin{aligned} C &= \frac{4g_1^2}{\kappa\gamma_1} = \frac{4g_2^2}{\kappa\gamma_2} \\ \gamma_{\text{sc}} &= \frac{\gamma_1 g_1^2|\alpha|^2}{\Delta^2} + \frac{\gamma_1 g_2^2|\alpha|^2}{\Delta^2} + \frac{\gamma_2 g_1^2|\alpha|^2}{\Delta^2} + \frac{\gamma_2 g_2^2|\alpha|^2}{\Delta^2} = \frac{(\gamma_1+\gamma_2)(g_1^2+g_2^2)|\alpha|^2}{\Delta} \\ p &= \frac{\gamma_1 g_2^2 + \gamma_2 g_1^2}{(\gamma_1+\gamma_2)(g_1^2+g_2^2)} = \frac{2\gamma_1\gamma_2}{(\gamma_1+\gamma_2)^2} \end{aligned} \tag{S9}$$

The cooperativity C is a property of cavity geometry. Its existence implies that $g_2^2/\gamma_2 = g_1^2/\gamma_1$ and hence that the rates of the $\hat{\sigma}^i_{\uparrow\downarrow}$ and $\hat{\sigma}^i_{\downarrow\uparrow}$ processes are equal. The parameter γ_{sc} is the total scattering rate of photons off the drive and is the sum of the rates of all single particle processes, and p is the spin flip probability, which is defined as the ratio between the sum of the rates of the $\hat{\sigma}^i_{\uparrow\downarrow}$ and $\hat{\sigma}^i_{\downarrow\uparrow}$ processes and $\gamma_{\rm sc}$. We have also made a specific choice of homodyne angle, $\phi = 2\arctan(2\delta/\kappa) - \pi$, to guarantee maximum measurement strength. To simplify notation, we define

$$\chi = \Big(\frac{(g_1^2+g_2^2)|\alpha|}{\Delta}\Big)^2 \frac{\delta_*}{\delta_*^2+(\kappa/2)^2} = \frac{C\gamma_{\rm sc} d/2}{d^2+1}, \qquad \Gamma = \Big(\frac{(g_1^2+g_2^2)|\alpha|}{\Delta}\Big)^2 \frac{\kappa}{\delta_*^2+(\kappa/2)^2} = \frac{C\gamma_{\rm sc}}{1+d^2}, \tag{S10}$$

and we have decided to express the effective parameters χ and Γ in terms of C, $\gamma_{\rm sc}$ and $d = 2\delta_*/\kappa$. Omitting single particle rotation terms, we can thus express the evolution of the system in terms of the effective equation

$$\begin{aligned} d\hat{\rho}_d = &-i\Big[\chi\hat{S}_z^2, \hat{\rho}_d\Big]\,dt + \Gamma\Big(\hat{S}_z\hat{\rho}_d\hat{S}_z - \frac{\{\hat{S}_z^2,\hat{\rho}\}}{2}\Big)\,dt + \frac{\gamma_{\rm sc}p}{2}\sum_{i=1}^N \Big(\hat{\sigma}^i_{\uparrow\downarrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\downarrow\uparrow} + \hat{\sigma}^i_{\downarrow\uparrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\uparrow\downarrow} - \hat{\rho}_d\Big)\,dt \\ &+ \frac{\gamma_{\rm sc}(1-p)}{2}\sum_{i=1}^N \Big(\hat{\sigma}^i_{\uparrow\uparrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\uparrow\uparrow} + \hat{\sigma}^i_{\downarrow\downarrow}\,\hat{\rho}_d\,\hat{\sigma}^i_{\downarrow\downarrow} - \hat{\rho}_d\Big)\,dt + \sqrt{\Gamma\eta}\Big(\hat{S}_z\hat{\rho}_d + \hat{\rho}_d\hat{S}_z - 2\langle\hat{S}_z\rangle\hat{\rho}_d\Big)\,dW. \end{aligned} \tag{S11}$$

Equations (S11) and (S10) are the same as Eq. (1) and Eq. (5) in the main text once we replace $\hat{\sigma}_{\uparrow\downarrow} = \hat{\sigma}^+$, $\hat{\sigma}_{\downarrow\uparrow} = \hat{\sigma}^-$, $\hat{\sigma}_{\uparrow\uparrow} = (1+\hat{\sigma}_z)/2$ and $\hat{\sigma}_{\downarrow\downarrow} = (1-\hat{\sigma}_z)/2$

We also summarize here the assumptions that went into this derivation. First $\Delta \gg \gamma, g_{1,2}\sqrt{N}, g_{1,2}|\alpha|$ ($|\alpha|^2$ is the number of photons in the cavity) for the elimination of the the excited state. Then $|\alpha|^2 \gg \langle\hat{b}^\dagger\hat{b}\rangle$ to treat the resulting dispersive atom-light interaction perturbatively during adiabatic elimination of the cavity. This translates into

$$\frac{(g_1^2+g_2^2)\sqrt{\langle\hat{S}_z^2\rangle}}{\Delta\sqrt{\delta_*^2+(\kappa/2)^2}} < \frac{2(g_1^2+g_2^2)\sqrt{\langle\hat{S}_z^2\rangle}}{\Delta\kappa} \ll 1 \to C\sqrt{\langle\hat{S}_z^2\rangle} \ll \frac{\Delta}{(\gamma_1+\gamma_2)/2}. \tag{S12}$$

We have considered the worst case scenario with $\delta_* = 0$ since this is relevant for QND measurements. For the initial conditions we are considering, $\langle\hat{S}_z^2\rangle \sim N$, so this assumption becomes $NC \ll \Delta\sqrt{N}/(\gamma_1+\gamma_2)$. A violation of this condition indicates that quantum fluctuations are strong enough that the relation between the adiabatically eliminated $\hat{b}$ and $\hat{S}_z$ can no longer be captured by the simple linear relation Eq. (S7) and a nonlinear approximation is required [S3].

LARGE N EQUATIONS FOR FIRST AND SECOND ORDER CORRELATORS

Here we compute the evolution equations for relevant observables. We calculate things exactly as much as possible, and apply approximations only in the end. The equations for linear observables are obtained by multiplying Eq. (1) with the appropriate operator and tracing the result. First we do averages of $\hat{S}_{x,y,z}$

$$\begin{aligned} d\langle\hat{S}_x\rangle &= \Big[-\chi\langle\{\hat{S}_z,\hat{S}_y\}\rangle - \frac{\Gamma}{2}\langle\hat{S}_x\rangle - \frac{\gamma_{\rm sc}}{2}\langle\hat{S}_x\rangle\Big]\,dt + \sqrt{\Gamma\eta}\Big[\langle\{\hat{S}_x,\hat{S}_z\}\rangle - 2\langle\hat{S}_x\rangle\langle\hat{S}_z\rangle\Big]\,dW \\ d\langle\hat{S}_y\rangle &= \Big[\chi\langle\{\hat{S}_z,\hat{S}_x\}\rangle - \frac{\Gamma}{2}\langle\hat{S}_y\rangle - \frac{\gamma_{\rm sc}}{2}\langle\hat{S}_y\rangle\Big]\,dt + \sqrt{\Gamma\eta}\Big[\langle\{\hat{S}_y,\hat{S}_z\}\rangle - 2\langle\hat{S}_y\rangle\langle\hat{S}_z\rangle\Big]\,dW \\ d\langle\hat{S}_z\rangle &= -\gamma_{\rm sc}p\langle\hat{S}_z\rangle\,dt + 2\sqrt{\Gamma\eta}\Big[\langle\hat{S}_z^2\rangle - \langle\hat{S}_z\rangle\langle\hat{S}_z\rangle\Big]\,dW, \end{aligned} \tag{S13}$$

where $\{\}$ is the anticommutator. Relevant two-point correlators are

$$\begin{aligned} d\langle\hat{S}_z^2\rangle &= -2\gamma_{\rm sc}p\Big[\langle\hat{S}_z\rangle^2 - \frac{N}{4}\Big]\,dt + 2\sqrt{\Gamma\eta}\Big[\langle\hat{S}_z^3\rangle - \langle\hat{S}_z^2\rangle\langle\hat{S}_z\rangle\Big]\,dW, \\ d\langle\{\hat{S}_z,\hat{S}_y\}\rangle &= \Big[2\chi\langle\{\hat{S}_z,\{\hat{S}_z,\hat{S}_x\}\}\rangle - \frac{\Gamma}{2}\langle\{\hat{S}_z,\hat{S}_y\}\rangle - \gamma_{\rm sc}(p+1/2)\langle\{\hat{S}_z,\hat{S}_y\}\rangle\Big]\,dt \\ &\quad + \sqrt{\Gamma\eta}\Big[\langle\{\hat{S}_z,\{\hat{S}_z,\hat{S}_y\}\}\rangle - 2\langle\{\hat{S}_z,\hat{S}_y\}\rangle\langle\hat{S}_z\rangle\Big]\,dW \\ d\langle\hat{S}_y^2\rangle &= \Big[\chi\langle\{\hat{S}_y,\{\hat{S}_z,\hat{S}_x\}\}\rangle + \Gamma\langle\hat{S}_x^2\rangle - \Gamma\langle\hat{S}_y^2\rangle - \gamma_{\rm sc}\Big(\langle\hat{S}_y^2\rangle - N/4\Big)\Big]\,dt \\ &\quad + \sqrt{\Gamma\eta}\Big[\langle\{\hat{S}_z,\hat{S}_y^2\}\rangle - 2\langle\hat{S}_z\rangle\langle\hat{S}_y^2\rangle\Big]\,dW \end{aligned} \tag{S14}$$

We can obtain exact equations for variances and covariances, keeping in mind the product rule $d(ab) = adb+bda+da\,db$ and the Ito rules $dW^2 = dt$, $dtdW = dt^2 = 0$ to get the right stochastic equations. Thus

$$\begin{aligned} d\,\mathrm{Var}(\hat{S}_z) &= \Big[-4\Gamma\eta\,\mathrm{Var}(\hat{S}_z)^2 - 2\gamma_{\mathrm{sc}}p\Big(\mathrm{Var}(\hat{S}_z) - \frac{N}{4}\Big)\Big]\,dt + 2\sqrt{\Gamma\eta}\Big[\langle \hat{S}_z^3\rangle - 3\langle \hat{S}_z^2\rangle\langle \hat{S}_z\rangle + 2\langle \hat{S}_z\rangle^3\Big]\,dW \\ d\,\mathrm{Cov} &= \Bigg(\frac{\chi}{2}\Big[\langle\{\{\hat{S}_x,\hat{S}_z\},\hat{S}_z\}\rangle - 2\langle \hat{S}_z\rangle\langle\{\hat{S}_x,\hat{S}_z\}\rangle\Big] - 4\Gamma\eta\mathrm{Var}(\hat{S}_z)\,\mathrm{Cov} - \Big[\frac{\Gamma}{2} + \gamma_{\mathrm{sc}}(p+1/2)\Big]\mathrm{Cov}\Bigg)\,dt \\ &\quad + \frac{\sqrt{\Gamma\eta}}{2}\Big[\langle\{\hat{S}_z,\{\hat{S}_y,\hat{S}_z\}\}\rangle - 4\langle\{\hat{S}_z,\hat{S}_y\}\rangle\langle\hat{S}_z\rangle - 4\langle\hat{S}_y\rangle\langle\hat{S}_z^2\rangle + 8\langle\hat{S}_y\rangle\langle\hat{S}_z\rangle^2\Big]\,dW \\ d\,\mathrm{Var}(\hat{S}_y) &= \Bigg(\chi\Big[\langle\{\hat{S}_y,\{\hat{S}_z,\hat{S}_x\}\}\rangle - 2\langle\hat{S}_y\rangle\langle\{\hat{S}_z,\hat{S}_x\}\rangle\Big] + \Gamma\Big[\langle\hat{S}_x^2\rangle - \mathrm{Var}(\hat{S}_y)\Big] - \gamma_{\mathrm{sc}}\Big[\mathrm{Var}(\hat{S}_y) - N/4\Big] - 4\Gamma\eta\,\mathrm{Cov}^2\Bigg)\,dt \\ &\quad + \sqrt{\Gamma\eta}\Big[\langle\{\hat{S}_y^2,\hat{S}_z\}\rangle - 2\langle\hat{S}_z\rangle\langle\hat{S}_y^2\rangle - 2\langle\hat{S}_y\rangle\langle\{\hat{S}_y,\hat{S}_z\}\rangle + 4\langle\hat{S}_y\rangle^2\langle\hat{S}_z\rangle\Big]\,dW, \end{aligned} \tag{S15}$$

where Var and $\mathrm{Cov} = \langle\{\hat{S}_z,\hat{S}_y\}\rangle/2 - \langle\hat{S}_y\rangle\langle\hat{S}_z\rangle$ are short-hands for variance and the YZ covariance, respectively. We now begin with the large N approximation. To apply it, we define the mean field equations of motion, obtained by replacing $(\langle\hat{S}_x\rangle,\langle\hat{S}_y\rangle,\langle\hat{S}_z\rangle) \to N(X,Y,Z)/2$ and $(\langle\hat{S}_z^2\rangle,\langle\{\hat{S}_z,\hat{S}_y\}\rangle,\langle\{\hat{S}_z,\hat{S}_x\}\rangle) \to N^2(Z^2,2YZ,2XZ)/4$ (factorization), where X,Y,Z are functions of time. This results in

$$\dot{X} = -\chi NYZ - \frac{\gamma_{\mathrm{sc}}}{2}X, \qquad \dot{Y} = \chi NXZ - \frac{\gamma_{\mathrm{sc}}}{2}Y, \qquad \dot{Z} = -\gamma_{\mathrm{sc}}pZ, \tag{S16}$$

where we have further neglected $\Gamma \ll \Gamma N$, since it is of higher order in the $1/N$ expansion. By keeping γ_{sc} we are implicitly assuming that $\gamma_{\mathrm{sc}} > \Gamma$. This is not necessarily always the case, especially when the single particle cooperativity $C \sim 1$. However, in all cases γ_{sc} is the obstruction to squeezing, and so we will always keep it. The mean field equations, together with initial conditions $X(0) = 1, Y(0) = Z(0) = 0$ result in $X = e^{-\gamma_{\mathrm{sc}}t/2}, Y = Z = 0$. We define the deviations from mean field $(\delta\hat{S}_x,\delta\hat{S}_y,\delta\hat{S}_z) = (\hat{S}_x - Ne^{-\gamma_{\mathrm{sc}}t}/2,\hat{S}_y,\hat{S}_z)$. In the large N limit, fluctuations and standard deviations are typically of size $\sqrt{N}$, so we further define $[v_{zz},v_{zy},v_{yy}] = 4[\mathrm{Var}(\hat{S}_z),\mathrm{Cov},\mathrm{Var}(\hat{S}_y)]/N$ and $(\hat{x},\hat{y},\hat{z}) = 2(\delta\hat{S}_x,\delta\hat{S}_y,\delta\hat{S}_z)/\sqrt{N}$. Initially, all of these quantities are ~ 1, so we expect that this will remain so for some time. In terms of these definitions (but still working exactly),

$$\begin{aligned} d\langle\hat{x}\rangle &= \Big[-\frac{\chi\sqrt{N}}{2}\langle\{\hat{z},\hat{y}\}\rangle - \frac{\Gamma\sqrt{N}e^{-\gamma_{\mathrm{sc}}t/2}}{2} - \frac{\Gamma}{2}\langle\hat{x}\rangle - \frac{\gamma_{\mathrm{sc}}}{2}\langle\hat{x}\rangle\Big]\,dt + \frac{\sqrt{\Gamma N\eta}}{2}\Big[\langle\{\hat{x},\hat{z}\}\rangle - 2\langle\hat{x}\rangle\langle\hat{z}\rangle\Big] \\ d\langle\hat{y}\rangle &= \Big[\chi Ne^{-\gamma_{\mathrm{sc}}t/2}\langle\hat{z}\rangle - \frac{\gamma_{\mathrm{sc}}}{2}\langle\hat{y}\rangle + \frac{\chi\sqrt{N}}{2}\langle\{\hat{z},\hat{x}\}\rangle - \frac{\Gamma}{2}\langle\hat{y}\rangle\Big]\,dt + \sqrt{\Gamma N\eta}\,v_{zy}\,dW \\ d\langle\hat{z}\rangle &= -\gamma_{\mathrm{sc}}p\,\langle\hat{z}\rangle\,dt + \sqrt{\Gamma N\eta}\,v_{zz}\,dW \end{aligned} \tag{S17}$$

and

$$\begin{aligned} dv_{zz} &= \Big[-\Gamma N\eta v_{zz}^2 - 2\gamma_{\mathrm{sc}}p(v_{zz}-1)\Big]\,dt + \sqrt{\Gamma N\eta}\Big[\langle\hat{z}^3\rangle - 3\langle\hat{z}^2\rangle\langle\hat{z}\rangle + 2\langle\hat{z}\rangle^3\Big]\,dW \\ dv_{zy} &= \Bigg(\chi Ne^{-\gamma_{\mathrm{sc}}t/2}v_{zz} - \Gamma Nv_{zz}v_{zy} - \gamma_{\mathrm{sc}}(p+1/2)v_{zy} + \frac{\chi\sqrt{N}}{4}\Big[\langle\{\{\hat{x},\hat{z}\},\hat{z}\}\rangle - 2\langle\hat{z}\rangle\langle\{\hat{x},\hat{z}\}\rangle\Big] - \frac{\Gamma}{2}v_{zy}\Bigg)\,dt \\ &\quad + \frac{\sqrt{\Gamma\eta N}}{4}\Big[\langle\{\hat{z},\{\hat{y},\hat{z}\}\}\rangle - 4\langle\{\hat{z},\hat{y}\}\rangle\langle\hat{z}\rangle - 4\langle\hat{y}\rangle\langle\hat{z}^2\rangle + 8\langle\hat{y}\rangle\langle\hat{z}\rangle^2\Big]\,dW \\ dv_{yy} &= \Bigg(2\chi Ne^{-\gamma_{\mathrm{sc}}t/2}v_{zy} + \Gamma Ne^{-\gamma_{\mathrm{sc}}t} - \gamma_{\mathrm{sc}}(v_{yy}-1) - \Gamma N\eta v_{zy}^2 \\ &\quad + \frac{\chi\sqrt{N}}{2}\Big[\langle\{\hat{y},\{\hat{z},\hat{x}\}\}\rangle - 2\langle\hat{y}\rangle\langle\{\hat{z},\hat{x}\}\rangle\Big] + 2\Gamma\sqrt{N}e^{-\gamma_{\mathrm{sc}}t/2}\langle\hat{x}\rangle + \Gamma\langle\hat{x}\rangle^2 - \Gamma\langle\hat{y}^2\rangle + \Gamma\langle\hat{y}\rangle^2\Bigg)\,dt \\ &\quad + \frac{\sqrt{\Gamma\eta N}}{2}\Big[\langle\{\hat{y}^2,\hat{z}\}\rangle - 2\langle\hat{z}\rangle\langle\hat{y}^2\rangle - 2\langle\hat{y}\rangle\langle\{\hat{y},\hat{z}\}\rangle + 4\langle\hat{y}\rangle^2\langle\hat{z}\rangle\Big]\,dW \end{aligned} \tag{S18}$$

Note that $\langle\hat{z}\rangle$ and $\langle\hat{y}\rangle$ are not zero since they are fluctuations with respect to the mean field values, not with respect to the true mean values of $\hat{z},\hat{y}$. The reason for this choice is that it makes the relative N-dependent size of various quantities more transparent. So far Eq. (S17) and Eq. (S18) are exact given that they are just rewritings. In the large

N limit we omit terms $\propto \chi\sqrt{N}, \Gamma\sqrt{N}, \Gamma \ll \chi N, \Gamma N$. The resulting evolution equations preserve gaussianity. Since the initial state is gaussian, the stochastic terms in the equations for v_{zz}, v_{zy}, v_{yy} drop out. We are thus led to

$$
\begin{aligned}
\dot{v}_{zz} &= -\Gamma N\eta v_{zz}^2 - 2\gamma_{\rm sc}p(v_{zz}-1) & dz &= -\gamma_{\rm sc}pz dt + \sqrt{\Gamma N\eta}v_{zz}\, dW \\
\dot{v}_{yz} &= \chi N e^{-\gamma_{\rm sc}t/2} - \Gamma N v_{zz}v_{zy} - \gamma_{\rm sc}(p+1/2)v_{zy} & dy &= \left(\chi N e^{-\gamma_{\rm sc}t/2}z - \frac{\gamma_{\rm sc}}{2}y\right)dt + \sqrt{\Gamma N\eta}v_{zy}\, dW \\
\dot{v}_{yy} &= 2\chi N e^{-\gamma_{\rm sc}t/2}v_{zy} + \Gamma N e^{-\gamma_{\rm sc}t} - \Gamma N\eta v_{zy}^2 - \gamma_{\rm sc}(v_{yy}-1), & &
\end{aligned}
\tag{S19}
$$

which are Eq. (2) and Eq. (3) in the main text, with $z = \langle\hat{z}\rangle$, $y = \langle\hat{y}\rangle$ and $\langle\hat{S}_x\rangle = \frac{N}{2}e^{-\gamma_{\rm sc}t/2}$. At the same time, the measurement record becomes

$$
dq = 2\sqrt{\Gamma\eta}\langle\hat{S}_z\rangle\, dt + dW = \sqrt{\Gamma N\eta}\, z\, dt + dW \tag{S20}
$$

Spin squeezing from second order correlators

To leading order in N, the Bloch vector is pointing along $+x$ with length $Ne^{-\gamma_{\rm sc}t/2}/2$. We thus need to consider fluctuations along the y and z directions. To be more specific, we define the matrix

$$
\begin{pmatrix} v_{zz} & v_{zy} \\ v_{zy} & v_{yy} \end{pmatrix}. \tag{S21}
$$

The minimal and maximal variances are, respectively, the smallest and largest eigenvalues of this matrix:

$$
v_{\rm min} = \frac{v_{zz}+v_{yy}}{2} - \sqrt{\frac{(v_{zz}-v_{yy})^2}{4} + v_{zy}^2}, \qquad v_{\rm max} = \frac{v_{zz}+v_{yy}}{2} + \sqrt{\frac{(v_{zz}-v_{yy})^2}{4} + v_{zy}^2}, \tag{S22}
$$

and the normalization we have chosen for v_{zz}, v_{yy}, v_{zy} ensures that a spin coherent state has $v_{\rm min} = v_{\rm max} = 1$. The naive state area is defined as the square root of the product of minimum and maximum variances. Thus

$$
A^* = \sqrt{v_{\rm min}\times v_{\rm max}} = \sqrt{v_{zz}v_{yy} - v_{zy}^2}. \tag{S23}
$$

The Wineland squeezing parameter and normalized state area are defined by dividing by the Bloch vector length (normalized by $N/2$) squared. Thus

$$
\xi^2 = \frac{v_{\rm min}}{e^{-\gamma_{\rm sc}t}} = e^{\gamma_{\rm sc}t}\left[\frac{v_{zz}+v_{yy}}{2} - \sqrt{\frac{(v_{zz}-v_{yy})^2}{4} + v_{zy}^2}\right], \qquad A = \frac{\sqrt{v_{\rm min}\times v_{\rm max}}}{e^{-\gamma_{\rm sc}t}} = e^{\gamma_{\rm sc}t}\sqrt{v_{zz}v_{yy} - v_{zy}^2}, \tag{S24}
$$

which is Eq. (6).

OPTIMAL SQUEEZNG WITH QND

With spin flips

The equations for v_{zz} and z are decoupled from the rest and can be solved exactly. We write the solution in the limit that $\Gamma N\eta \gg \gamma_{\rm sc}p$. Then

$$
\begin{aligned}
\dot{v}_{zz} &= -\Gamma N\eta\left(v_{zz} + \frac{\gamma_{\rm sc}p}{\Gamma N\eta} + 2\sqrt{\frac{\gamma_{\rm sc}p}{2\Gamma N\eta}}\sqrt{1-\frac{\gamma_{\rm sc}p}{2\Gamma N\eta}}\right)\left(v_{zz} + \frac{\gamma_{\rm sc}p}{\Gamma N\eta} - 2\sqrt{\frac{\gamma_{\rm sc}p}{2\Gamma N\eta}}\sqrt{1-\frac{\gamma_{\rm sc}p}{2\Gamma N\eta}}\right) \\
&\approx -\Gamma N\eta\left(v_{zz} + \sqrt{2\frac{\gamma_{\rm sc}p}{\Gamma N\eta}}\right)\left(v_{zz} - \sqrt{\frac{2\gamma_{\rm sc}p}{\Gamma N\eta}}\right).
\end{aligned}
\tag{S25}
$$

We want to solve z in terms of the measurement record, which is the accesible information, not in terms of dW so we rewrite its equation as $dz = -(\Gamma\eta N v_{zz} + \gamma_{\rm sc}p)z\, dt + \sqrt{\Gamma N\eta}\, v_{zz}\, dq$. The solutions for v_{zz} and z (with $v_{zz}(0) = 1$ and $z(0) = 0$) are then

$$
v_{zz}(t) \approx \epsilon\coth\left(\frac{\Gamma N\eta t + 1}{\Gamma N\eta\tau}\right), \qquad z(t) \approx \frac{\sqrt{2\gamma_{\rm sc}p}}{\sinh\left[\frac{\Gamma N\eta t+1}{\Gamma N\eta\tau}\right]}\int_0^t \cosh\left[\frac{\Gamma N\eta s + 1}{\Gamma N\eta\tau}\right] dq(s), \tag{S26}
$$

where $\epsilon = \sqrt{2\gamma_{\rm sc}p/(\Gamma N\eta)}$ and $\tau^{-1} = 2\sqrt{\gamma_{\rm sc}\Gamma N\eta\gamma_{\rm sc}p}$. Note that $\Gamma N\eta\tau = 1/\epsilon \gg 1$. When $t \ll \tau$ spin flips are not active, and the solutions reduce to $v_{zz}(t) = (1+\Gamma N\eta t)^{-1}$ and $z(t) \approx \sqrt{\Gamma N\eta}(1+\Gamma N\eta t)^{-1}\int_0^t dq(s)$, which is the time-averaged measurement record, in accordance with well-known results. In the opposite limit, when $t \gtrsim \tau$ and spin flips are active, we get $v_{zz}(t) \approx \sqrt{\epsilon}(1+2e^{-2t/\tau})$ and $z(t) \approx \sqrt{2\gamma_{\rm sc}p}\int_0^t e^{-(t-s)/\tau}\,dq(s)$. Thus, the z deflection of the Bloch vector is determined by the measurement record only within a time-window of size τ. Spin flips have effectively introduced a memory time.

Without spin flips

In this case ($p=0$), Eq. (1) can be solved exactly (they can also be solved exactly for nonzero χ, but this is not relevant for the QND analysis). The first thing to notice is that when $p=0$ the single particle dephasing terms commute (in the superoperator sense) with the rest of the evolution equation. We thus define $\hat{\upsilon} = e^{-\mathcal{L}_{\rm deph}t}\hat{\rho}$, where $\mathcal{L}_{\rm deph} = \sum_i(\mathcal{L}_{\hat{\sigma}^i_{\uparrow\uparrow}} + \mathcal{L}_{\hat{\sigma}^i_{\downarrow\downarrow}})$ so that

$$\mathrm{Tr}(\hat{S}_z\hat{\rho}) = \mathrm{Tr}(\hat{S}_z\hat{\upsilon}), \qquad \mathrm{Tr}(\hat{S}_z^2\hat{\rho}) = \mathrm{Tr}(\hat{S}_z^2\hat{\upsilon}), \qquad \mathrm{Tr}(\hat{S}_x\hat{\rho}) = e^{-\gamma_{\rm sc}t/2}\mathrm{Tr}(\hat{S}_x\hat{\upsilon}), \tag{S27}$$

and $\hat{\upsilon}$ satisfies

$$d\hat{\upsilon} = \Gamma\Big(\hat{S}_z\hat{\upsilon}\hat{S}_z - \frac{\{\hat{S}_z^2,\hat{\upsilon}\}}{2}\Big)\,dt + \sqrt{\Gamma\eta}\Big(\hat{S}_z\hat{\upsilon} + \hat{\upsilon}\hat{S}_z - 2\langle\hat{S}_z\rangle\Big)\,dW, \tag{S28}$$

and the expectation value is now taken with respect to $\hat{\upsilon}$. The matrix element $\langle m|\hat{\upsilon}|n\rangle$ satisfies the equation

$$d\langle m|\hat{\upsilon}|n\rangle = -\frac{\Gamma(m-n)^2}{2}\langle m|\hat{\upsilon}|n\rangle\,dt + \sqrt{\Gamma\eta}(m+n)\langle m|\hat{\upsilon}|n\rangle\,dW - 2\sqrt{\Gamma\eta}\langle\hat{S}_z\rangle\langle m|\hat{\upsilon}|n\rangle\,dW, \tag{S29}$$

which can be integrated taking into account the Ito rule. Up to m and n independent prefactors, this yields

$$\langle m|\hat{\upsilon}|n\rangle(t) \propto e^{-\Gamma t(m-n)^2/2}\, e^{-\Gamma\eta t(n+m)^2/2}\, e^{\sqrt{\Gamma\eta}(m+n)\left(W_t + 2\sqrt{\Gamma\eta}\int_0^t \langle\hat{S}_z\rangle\,dt'\right)} e^{-2\sqrt{\Gamma\eta}\int_0^t \langle\hat{S}_z\rangle\,dW_{t'}}\, \overbrace{e^{-(m^2+n^2)/N}}^{\langle m|\hat{\upsilon}|n\rangle(0)}, \tag{S30}$$

where we have approximated $\langle m|\hat{\upsilon}|n\rangle(0)$ and $W_t = \int_0^t dW$ is a Brownian process. With this explicit form for the matrix elements, we can calculate the relevant expectation values

$$\begin{aligned}
\mathcal{C} &= \sum_{n=-\infty}^{\infty} e^{-2\left(\frac{1}{N}+\Gamma\eta t\right)(n-n^*)^2}, &\qquad \mathrm{Tr}(\hat{\rho}\hat{S}_x) &= \frac{Ne^{-\frac{\Gamma t}{2}-\gamma_{\rm sc}t/2}}{2\mathcal{C}}\sum_{n=-\infty}^{\infty} e^{-2\left(\frac{1}{N}+\Gamma\eta t\right)(n-n^*+\frac{1}{2})^2}\\
\mathrm{Tr}(\hat{\rho}\hat{S}_z) &= \frac{1}{\mathcal{C}}\sum_{n=-\infty}^{\infty} e^{-2\left(\frac{1}{N}+\Gamma\eta t\right)(n-n^*)^2}n, &\qquad \mathrm{Tr}(\hat{\rho}\hat{S}_z^2) &= \frac{1}{\mathcal{C}}\sum_{n=-\infty}^{\infty} e^{-2\left(\frac{1}{N}+\Gamma\eta t\right)(n-n^*)^2}n^2
\end{aligned} \tag{S31}$$

where we have extended the sums to $\pm\infty$ and

$$n^* \equiv \frac{\sqrt{\Gamma\eta}/2}{\Gamma\eta t + 1/N}\left(W_t + 2\sqrt{\Gamma\eta}\int_0^t \langle\hat{S}_z\rangle\,dt'\right) \tag{S32}$$

sets the average $\langle\hat{S}_z\rangle$ and thus satisfies the consistency requirement

$$dn^* = \frac{\Gamma\eta}{\Gamma\eta t + \frac{1}{N}}\left[\frac{\sum_{n=-\infty}^{\infty} e^{-2\left(\frac{1}{N}+\Gamma\eta t\right)(n-n^*)^2}(n-n^*)}{\sum_{n=-\infty}^{\infty} e^{-2\left(\frac{1}{N}+\Gamma\eta t\right)(n-n^*)^2}}\right]dt + \frac{\sqrt{\Gamma\eta}}{2\left(\Gamma\eta t + \frac{1}{N}\right)}dW_t. \tag{S33}$$

Average quantities are then computed by first calculating them at fixed n^* and then sampling n^* from the previous stochastic equation. When $\Gamma\eta\, t \lesssim 1$, the arguments of the sums vary slowly and can be approximated by integrals, leading to

$$\xi^2 = \frac{e^{(\Gamma+\gamma_{\rm sc})t}}{1+\Gamma N\eta t} \longrightarrow \Gamma t_{\rm opt} = \frac{1}{1+\frac{\gamma_{\rm sc}}{\Gamma}} = \frac{C}{C+1}, \qquad \xi^2_{\rm opt} = \frac{e}{\eta N}\left(1+\frac{1}{C}\right). \tag{S34}$$

Note that $\Gamma t_{\rm opt} < 1$ always so that $\Gamma\eta t_{\rm opt}$ satisfies $\Gamma\eta t_{\rm opt} \lesssim 1$ quite generically, except when $\eta = 1$ and $C \gg 1$. Note also that squeezing is independent of n^* within this approximation. Beyond this regime, which is a situation relevant only when $C \gtrsim 1$ and $\eta \approx 1$, the result does depend on n^*. In this scenario, squeezing can reach $\xi^2 = 2/N$ on specific trajectories (when n^* is even) but the average over trajectories is still captured relatively well by Eq. (S34).

OPTIMAL SPIN SQUEEZING WITH OAT

When $\eta = 0$, Eqs. (S19) become linear. The solutions are $y = z = 0$ (no stochastic terms), $v_{zz} = 1$, and

$$\begin{aligned}
v_{zy} &= \frac{\chi N}{\gamma_{\rm sc} p} e^{-\gamma_{\rm sc} t}(1 - e^{-\gamma_{\rm sc} p t}) \\
v_{yy} &= 1 + \Gamma N t e^{-\gamma_{\rm sc} t} + \frac{2(\chi N)^2}{(\gamma_{\rm sc} p)^2} e^{-\gamma_{\rm sc} t}(\gamma_{\rm sc} p t - 1 + e^{-\gamma_{\rm sc} p t}) \\
A^2 &= e^{2\gamma_{\rm sc} t}(v_{yy} v_{zz} - v_{zy}^2) = e^{2\gamma_{\rm sc} t}\left[1 + \Gamma N t e^{-\gamma_{\rm sc} t} + \frac{2(\chi N)^2}{(\gamma_{\rm sc} p)^2}\left(\gamma_{\rm sc} p t - \frac{3}{2} + 2e^{-\gamma_{\rm sc} p t} - \frac{e^{-2\gamma_{\rm sc} p t}}{2}\right)\right], \\
\xi^2 &= \frac{e^{\gamma_{\rm sc} t}(v_{zz} v_{yy} - v_{zy}^2)}{\frac{1+v_{yy}}{2} + \sqrt{\left(\frac{1-v_{yy}}{2}\right)^2 + v_{zy}^2}}
\end{aligned} \tag{S35}$$

where A is the state area and ξ^2 is the squeezing parameter.

With spin flips

We expect the optimal squeezing to occur for $\gamma_{\rm sc} t \ll 1$, so we calculate things perturbatively in $\gamma_{\rm sc}$. This leads to $v_{zy} \approx \chi N t$, $v_{yy} \approx 1 + \Gamma N t + \chi^2 N^2 t^2 \approx \chi^2 N^2 t^2$ ($\chi N t \gg 1$ since we would not have any squeezing otherwise). Then

$$\begin{aligned}
A^2 &\approx v_{zz} v_{yy} - v_{zy}^2 \approx 1 + \Gamma N t + \frac{2(\chi N)^2 \gamma_{\rm sc} t^3}{3} \\
\xi^2 &\approx \frac{A^2}{v_{yy}} = \frac{1 + \Gamma N t}{(\chi N t)^2} + \frac{2}{3}\gamma_{\rm sc} p t,
\end{aligned} \tag{S36}$$

in agreement with the exact treatment presented in Ref. [S4]. For smaller detunings $d = 2\delta_*/\kappa$, the collective dephasing term is active and the minimum squeezing is the result of the minimization of

$$\xi^2 \approx \frac{\Gamma/\chi}{\chi N t} + \frac{2}{3}\gamma_{\rm sc} p t \longrightarrow \xi_t^2 = \sqrt{\frac{32p}{3NC}}\sqrt{1 + \frac{1}{d^2}}, \qquad \gamma_{\rm sc} p\, t_{\rm opt} = \frac{3\xi_t^2}{4}, \qquad A^2 = 2\sqrt{6}\sqrt{\frac{NC}{pd^2(d^2+1)}} \tag{S37}$$

where we have used Eq. (5). This expression for squeezing leads to Eq. (9) for $d \gtrsim 1$. Note also that, assuming p is not too small, the optimal time $t_{\rm opt}$ satisfies $\gamma_{\rm sc} t_{\rm opt} \ll 1$, as promised. Even if ξ_t^2 is relatively insensitive to d at large values of d, the area A gets smaller at larger detunings, which is desirable. As d is further increased, Γ gets small very quickly, and the optimal squeezing is then the result of minimizing (assuming $d \gg 1$)

$$\xi^2 = \frac{1}{(\chi N t)^2} + \frac{2}{3}\gamma_{\rm sc} p t \longrightarrow \xi_t^2 = 3^{1/3}\left(\frac{2pd}{NC}\right)^{2/3}, \qquad \gamma_{\rm sc} p t_{\rm opt} = \xi_t^2, \qquad A^2 \approx 3 \tag{S38}$$

Squeezing gets worse as d increases, but the area is reasonably small, so it's a good idea to operate at the crossover between these two regimes. We estimate the detuning value at which this happens by equating the expressions for squeezing on both sides. This leads to

$$d_{\rm crossover} \approx \frac{8}{3^{5/4}}\left[\frac{NC}{2p}\right]^{1/4} \approx 2\left[\frac{NC}{2p}\right]^{1/4} \tag{S39}$$

No spin flips

We set $p = 0$ here, but do not assume $\gamma_{\rm sc} t \ll 1$ necessarily. Then $v_{yy} \approx (\chi N t)^2 e^{-\gamma_{\rm sc} t}$, $(A^*)^2 \approx v_{yy} \times v_{\rm min}$ and

$$v_{\rm min} \approx \frac{e^{\gamma_{\rm sc} t} + \Gamma N t}{(\chi N t)^2} + \underbrace{\frac{(\chi N t)^4}{6N^2}}_{\text{curvature}}, \tag{S40}$$

where the extra curvature term is a correction that is not captured by Eq. (S19) since it is of next to leading order in $1/N$. Furthermore, loss of contrast due to $\gamma_{\rm sc}$ may be relevant. With $\langle \hat{S}_x\rangle \approx Ne^{-\gamma_{\rm sc}t}/2$, the squeezing parameter is

$$\xi^2 = e^{\gamma_{\rm sc}t}\left[\frac{e^{\gamma_{\rm sc}t}+\Gamma N t}{(\chi N t)^2} + \frac{(\chi N t)^4}{6N^2}\right]. \tag{S41}$$

In fact, for $p=0$ and $\eta=0$, Eq. (1) can be solved exactly since all the terms in the master equation commute (in the superoperator sense). The exact values for relevant observables are

$$\begin{aligned}
\langle \hat{S}_z^2\rangle &= \frac{N}{4}\\
\langle\{\hat{S}_z,\hat{S}_y\}\rangle &= \frac{N(N-1)}{2}e^{-(\gamma_{\rm sc}+\Gamma)t/2}\cos(\chi t)^{N-2}\sin(\chi t)\\
\langle \hat{S}_y^2\rangle &= \frac{N}{4}(1-e^{-\gamma_{\rm sc}t}) + \frac{N(N+1)e^{-\gamma_{\rm sc}t}}{8} + \frac{N(1-N)e^{-\gamma_{\rm sc}t-2\Gamma t}}{8}\cos(2\chi t)^{N-2}\\
\langle \hat{S}_x\rangle &= e^{-(\gamma_{\rm sc}+\Gamma)t/2}\cos(\chi t)^{N-1}.
\end{aligned} \tag{S42}$$

A comparison of squeezing (optimized over time) obtained with the exact equations and with Eq. (S41) is shown in Fig. S1(a) for $N=10^6$ and $C=10^2$. Since the agreement is almost perfect, we keep working with Eq. (S41) for analytical insight.

As Fig. S1(a) demonstrates, there can be three regimes. At smaller detunings (but still $d\gtrsim 1$), collective dephasing is active but single particle dephasing is not. The optimal squeezing is then limited by curvature and is obtained by minimizing

$$\xi^2 \approx \frac{\Gamma/\chi}{\chi N t} + \frac{(\chi N t)^4}{6N} \longrightarrow \xi_t^2 \approx \frac{5/2}{3^{1/5}d^{4/5}N^{2/5}},\quad \chi N t_{\rm opt} = \left(\frac{3N^2}{d}\right)^{1/5},\quad A\approx\sqrt{1+\frac{3.1N^{4/5}}{d^{6/5}}}. \tag{S43}$$

As the detuning is increased, both the optimal squeezing and the area improve. Eventually neither collective nor single particle dephasing will be active but the optimal squeezing will still be limited by curvature and is obtained by minimizing

$$\xi^2 \approx \frac{1}{(\chi N t)^2} + \frac{(\chi N t)^4}{6N^2} \longrightarrow \xi_t^2 \approx \frac{3^{2/3}/2}{N^{2/3}},\quad \chi N t_{\rm opt} = (3N^2)^{1/6},\quad A\approx\sqrt{1.5}. \tag{S44}$$

This is the well-known OAT result. Finally, at much larger detunings χ is small enough that curvature is irrelevant, and the main limitations are now single particle dephasing and contrast decay. Hence, the optimal squeezing is obtained by minimizing

$$\xi^2 = \frac{e^{2\gamma_{\rm sc}t}}{(\chi N t)^2} \longrightarrow \xi_t^2 \approx \left(\frac{ed}{NC}\right)^2,\quad \gamma_{\rm sc}t_{\rm opt} = 1,\quad A\approx\sqrt{e}. \tag{S45}$$

We can estimate the detunings at which the behaviour of ξ_t^2 changes by equating the values of the time optimized squeezing on both sides of a crossover. We find these detuning values to be

$$d_{c1} = \frac{5^{5/4}}{3^{13/12}}N^{1/3} \approx 2.3N^{1/3},\qquad d_{c2} = \frac{3^{1/3}N^{2/3}C}{e\sqrt{2}} \approx 0.4CN^{2/3} \tag{S46}$$

For the unitary region in the middle to exist, we need $d_{c1} < d_{c2}$, which requires that $C \gtrsim 6N^{-1/3}$. Otherwise, the two outer regions in Fig. merge, and the detuning optimized squeezing will be worse than the OAT result. We show the results for the area A in Fig. S1(b).

EXAMPLES

We illustrate our results using examples that are related to the experimental system we analyzed in the main text. We will consider various QND and OAT implementations, including systems that are not directly described by Eq. (1).

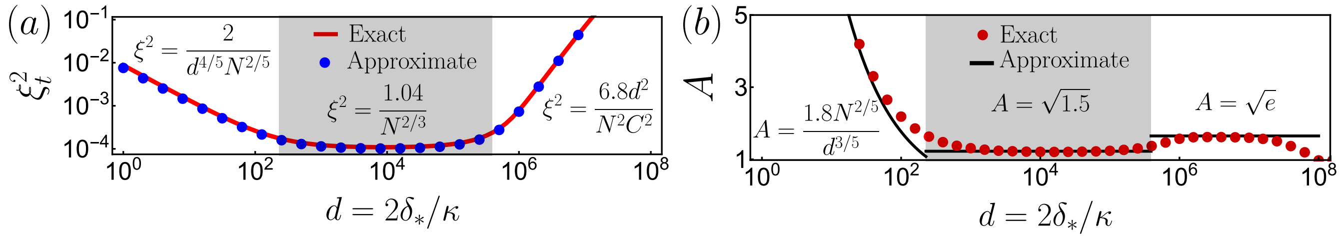

FIG. S1. (a) Comparison of time optimized ξ_t^2 using the exact Eqs. (S42) (solid red) and approximate Eq. (S41) (blue dots) as a function of detuning $d = 2\delta_*/\kappa$ for $N = 10^6$ and $C = 10^2$. Boundary between different shaded regions are given by $d_{c1} = 2.3N^{1/3}$ and $d_{c2} = 0.4CN^{2/3}$. (b) Area at the optimal squeezing time as a function of detuning. Red dots are calculated using Eqs. (S42), while black lines are the approximations for each of the regions described in section.

- We first analyze Ref. [S5], which implemented OAT using the same level scheme as the one in Fig. 1. Using $p = 1/2$, $C = 0.14$ and $N = 3 \times 10^4$ (see second column of page 2 of Ref. [S5], where single atom cooperativity is denoted η instead of C) leads to 14.5 dB of noise reduction according to Eq. (9). The experiment reports 10(1) dB of inferred noise reduction (after subtraction of detection noise, see first column of page 4) and is further affected by fluctuations in intracavity probe power and dephasing effects that reduce the effective radius of the Bloch vector. Importantly, a theoretical analysis of this experiment was presented in Refs. [S6, S7] and Eq. (9) is the same as the equation reported in Ref. [S7] in the presence of free space scattering.

- Next, we analyze Ref. [S8], which implemented QND measurements in the same level structure as Ref. [S5]. Using $p = 1/2$, $\eta = 0.4$ (quantum efficiency, see second column of page 2 of Ref. [S8]), and $NC = 3100 \times 3/2$ (see second column of page 3, the factor 3/2 comes from discrepancies in the definition of our C and the effective cooperativity in Ref. [S8]) leads to a noise reduction of 16 dB, according to Eq. (7) and 18 dB if $\eta = 1$, in agreement with the fundamental limit provided in Eq. (20) of the Supplementary Material of Ref. [S8]. Technical noise modifies this value to 9 dB of noise reduction and 3 dB of metrological enhancement when including contrast decay. As discussed in the Supplementary Material of Ref. [S8], in the ideal scenario the fundamental limit of 18 dB is not modified by the presence of contrast decay.

- Now we analyze Ref. [S9], which implemented QND measurements in the same level structure as Ref. [S8], but in a configuration where the cavity is closer to resonance with $|\uparrow\rangle$ than $|\downarrow\rangle$ and with different hyperfine levels. The different choice of levels means that $p \to 0$. In this QND setup, application of Eq. (13) with $\eta = 0.37$, $N = 4 \times 10^5$ and $C = 0.044$ leads to 34 dB of squeezing (without taking into account inhomogeneous couplings of the atoms to the cavity mode), compared to the 19 dB of noise reduction and 17 dB of metrological enhancement reported in Ref. [S9]. As discussed in the cited reference, detection noise (25 dB below quantum projection noise) and optomechanical effects were the main technical limitations.

- Then we analyze Ref. [S10], which does QND measurements in the same level structure and also the same levels as Ref. [S8]. Using $p = 1/2$, $\eta = 0.16$ and $NC = (5 \times 10^5)(0.78) \times 3/2$ (the factor of 3/2 arises because of discrepancies with our definition of cooperativity) leads, via Eq. (7), to 25 dB of squeezing. The cited reference reports 20 dB of metrological enhancement and predicts 24 dB. The difference of this prediction with our calculation originates from the broadening of the cavity linewidth due to free space scattering, a effect that we neglect and that arises as a higher order term in the adiabatic elimination of the excited state.

- Let us now study the QND implementation of Ref. [S11]. The setup involves $N \approx 10^{11}$ atoms in a vapour cell but has no cavity. To establish a comparison to our results, we thus need to find the quantity in Ref. [S11] analogous to the collective cooperativity C in terms of which we express all of our findings. In this case, it will be the optical depth of the atomic sample. We relate both by using the definition of C

$$NC = \frac{4g_1^2 N}{\kappa\gamma_1}, \tag{S47}$$

and expressing γ_1, g_1 and κ in terms of atomic and cavity properties. We use the Wigner-Weisskopf expression for γ_1, Eq. (2) of Ref. [S12] for g_1 and express κ in terms of the cavity's free spectral range and finesse $\mathcal{F}$:

$$\gamma_1 = \frac{\omega_1^3\mu^2}{3\pi\hbar\epsilon_0 c^3} \qquad g_1 = \left(\frac{\mu^2\omega_1}{2\hbar\epsilon_0 V_m}\right)^{1/2} \qquad \kappa = \frac{\pi c}{L\mathcal{F}}, \tag{S48}$$

where ω_1 is the transition frequency from $|\uparrow\rangle$ to $|e\rangle$, μ is the transition dipole moment, L is the cavity length and V_m is the mode volume. Putting all these together we have that

$$NC = \frac{6c^2}{\omega_1^2} \times \frac{N}{V_m} \times L \times \mathcal{F}. \tag{S49}$$

The resonant scattering cross section is $\sigma_1 = 4\pi c^2/\omega_1^2$, and we can take N/V_m as a representative of the average atom density n. Then

$$NC = \frac{3}{2\pi} n\sigma_1 L\mathcal{F} = \frac{3}{2\pi}(OD)\mathcal{F} \tag{S50}$$

where $OD = n\sigma_1 L$ is the optical depth of the sample. The experiment in Ref. [S11] works at an optical depth of 70 and uses no cavity, so we set $\mathcal{F} = 1$. Using these numbers to obtain a representative NC and using Eq. (7) with $\eta = 1$ (perfect detection efficiency) and $p = 1/2$ we obtain 7.5 dB as the limit to squeezing caused by scattering from the excited state. The experiment predicts 5.3 dB and reports 3.4 dB of noise reduction (see last paragraph of Supplementary Information of Ref. [S11] for a detailed discussion), which are close to our bound, and at the level where the differences in implementation (which may introduce factors of order 1) matter. Furthermore, at these smaller values of "NC" contrast decay modifies the attainable metrological enhancement by amounts that are sizeable relative to the noise reduction, and Ref. [S11] reports 2.3 dB of actual squeezing.

Finally, Ref. [S11] uses an experimental scheme put forward in Ref. [S13]. The Supplementary Material of Ref. [S13] also includes the limitations to squeezing coming from probe-induced scattering and arrive at Eq. (S40). Up to factors of order 1 this is the same as Eq. (7) in our paper, once the equivalence between NC, optical depth and finesse is made.

- The last experiment we analyze is the OAT implementation of Ref. [S14]. In this experiment, the cavity is parked close to resonance with one of the atomic transition frequencies and $p = 0$, similar to Ref. [S9], albeit using a different atom. For technical reasons, the experiment employs a two-tone scheme for the generation of squeezing. The first, blue-detuned, tone operates with an atom-drive detuning that falls in the center region of Fig. 4(a) while the second, red-detuned, tone is in the left region of Fig. 4(a) (numerical values for κ and the atom-drive detunings are given in sections I and V of the Supplementary Material of Ref. [S14]). For simplicity, we use the bound in the center region, which is the Kitagawa-Ueda limit. With $N = 1000$ this leads to 20 dB of squeezing, but inhomogeneous couplings of the atoms to the cavity mode lead to 16 dB, as discussed in section VIII of the Supplementary Material of Ref. [S14]. The experiment reports 16 dB of noise reduction, very much in agreement with the theoretical bound, and 12 dB of inferred metrological enhancement.

These considerations are summarized in Table I.

Experiment	Inferred noise reduction	Estimation by Ref.	Our prediction
Ref. [S5] (OAT)	10 dB	-	14.5 dB
Ref. [S8] (QND)	9 dB	18 dB	18 dB
Ref. [S9] (QND)	19 dB	-	34 dB
Ref. [S10] (QND)	20 dB	24 dB	25 dB
Ref. [S11] (QND)	3.4 dB	5.3 dB	7.5 dB
Ref. [S14] (OAT)	16 dB	16 dB	20 dB

TABLE I. Squeezing in experiments

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